

Shaheed Mohamed Ali

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Education

University of Waterloo
Bachelor of Computer Science

Waterloo, Ontario
Expected August 2027

Technical Skills

- **Languages:** Python, Java, JavaScript, TypeScript, Go, C++, C#, SQL
- **Frameworks:** React, Next.js, Node.js, FastAPI, Spring Boot, ASP.NET Core
- **Cloud & Reliability:** AWS, Azure, Docker, Kubernetes, Jenkins, Terraform, Helm, Linux, CI/CD, Unit Testing

Experience

DevOps Engineer Intern

Jun 2026 — Aug 2026

Proofpoint

Toronto, Ontario

- Cut **annual AWS spend by nearly \$100K** by replacing per-run EC2 Windows Server and Exchange environments with containerized Samba and Keycloak services on Kubernetes
- Cut **Jenkins build time by 45 minutes** by containerizing a Java Spring Boot test platform and automating builds and deployments with Terraform and Helm
- Eliminated **100% of per-run VM provisioning** by separating application, identity, and file services into reproducible Docker and Kubernetes components
- Consolidated **3 integration service layers into 1 reproducible CI environment** by packaging application, identity, and file dependencies as versioned containers

Software Engineer Intern

Sep 2025 — Dec 2025

FluidAI

Waterloo, Ontario

- Kept queries **under 500 ms across 16M records** by profiling MySQL execution plans and redesigning schemas with targeted indexes
- Reduced clinician response time by **60% with sub-second processing** by building an event-driven C# and Azure Cosmos DB detection pipeline
- Delivered **15+ production features supporting a \$2M hospital contract** by shipping Next.js and ASP.NET Core services with PostgreSQL, Docker, and Azure
- Consolidated **3 clinical contexts into 1 workflow** by building a LangChain assistant across patient history, symptoms, and postoperative status

Software Engineer Intern

Jan 2025 — May 2025

All Equip Repair & Service Ltd.

Stratford, Ontario

- Saved **\$96K annually** by solely developing a TypeScript, MongoDB, and Python workforce platform that replaced an \$8K-per-month HR system
- Saved HR **16 hours per week** by building a Python and SQL extraction pipeline that converted verified attendance into configurable timesheets and payslips

Software Engineer Intern

May 2024 — Aug 2024

Soilbrain Inc.

Waterloo, Ontario

- Reduced manual intervention by **90% across 50K+ daily IoT readings** by building a Python and AWS Lambda ingestion pipeline with health monitoring and automated alerts
- Improved data accuracy by **35% and cut latency by 40% across 10M+ monthly points** by debugging and optimizing C++ microcontroller transformation algorithms

Projects

Real-Time Network Packet Analyzer | Go, C++, Python, React, TypeScript

- Lowered packet-capture latency to **under 100 ms** by engineering concurrent Go and C++ processing, profiling sustained loads, and debugging memory faults and race conditions with GDB
- Rendered **50K+ packets in under 100 ms** by optimizing C++ parsing algorithms and data structures before streaming results to a React and TypeScript dashboard

Automated Document Search System | Python, LangChain, RAG, ChromaDB, FastAPI, Next.js

- Achieved **sub-second semantic search across 10K+ documents** by building a FastAPI ingestion and retrieval pipeline with LangChain, RAG, and ChromaDB
- Automated a **3-step document workflow** by orchestrating analysis, retrieval, and Typst compilation through tested FastAPI endpoints

Distributed Auction Processing Platform | C#, ASP.NET Core, PostgreSQL, Docker, RabbitMQ

- Sustained **bid processing under 100 ms across 3 services** with asynchronous RabbitMQ events, containerized boundaries, and unit-tested domain models
- Processed **5K+ auction transactions with 99.9% reliability** by validating C# service boundaries with unit and integration tests